

- Brushed 24V DC Motor

3.6 kg.cm / 46 RPM w/ 99.5:1 Planetary Gearbox

- Material - ABS

$$E = 290075.4753 \text{ psi} \rightarrow 1.99 \text{ GPa}$$

$$S_y = 44.8 \text{ MPa}$$

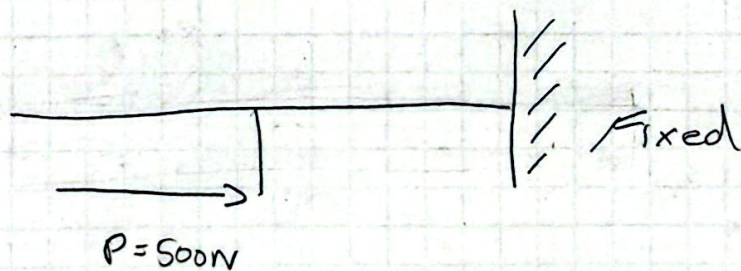
$$\text{Safety Factor} = 3$$

$$P = 300 \text{ N}$$

$$\sigma_{\text{allow}} = \frac{S_y}{n} = \frac{44.8}{3} = 14.93 \text{ MPa}$$

$$L = 18 + 36.6 = 54.6 \text{ mm}$$

Feature 1



Known

$$P = 300 \text{ N}$$

$$L_1 = 54.6 \text{ mm}$$

$$E = 2 \text{ GPa} = 2000 \text{ MPa}$$

$$S_y = 44.8 \text{ MPa}$$

$$n = 3 \quad b = 50 \text{ mm}$$

$$\delta_{\text{max}} = 0.30 \text{ mm}$$

Unknown

$$h =$$

$$R_A \quad M_A$$

$$\sum F_y = 0$$

$$R_A = P = 300 \text{ N}$$

$$\sum M_A = 0 = PL = 300(54.6) = 16,380 \text{ N} = 16.38 \text{ N}$$

$$I = \frac{bh^3}{12} \quad \sigma_{\text{max}} = \frac{Mc}{I} \quad c = \frac{h}{2}$$

$$\sigma_{\text{max}} = \frac{M(h/2)}{bh^3/12} = \boxed{\frac{6M}{bh^2}}$$

$$M = PL$$

$$\sigma_{\text{max}} = \frac{6PL}{bh^2}$$

$$\sigma_{\text{max}} \leq \frac{S_y}{n} \quad \frac{6PL}{bh^2} \leq \frac{S_y}{n}$$

$$h \geq \sqrt{\frac{6PLn}{bS_y}}$$

$$\delta = \frac{PL^3}{3EI} = \frac{PL^3}{3E(bh^3/12)}$$

$$\delta = \frac{4PL^3}{Ebh^3} = \boxed{h^3 \sqrt{\frac{4PL^3}{E b \delta}}}$$

$$h = \sqrt[3]{\frac{4(300)(54.6)^3}{(1990)(50)(0.30)}} = 18.7 \text{ mm}$$

$$h \geq \sqrt{\frac{6(300)(54.6)(3)}{(50)(44.9)}} = 11.1 \text{ mm}$$

$$h_{\text{stress}} = 11.1 \text{ mm}$$

$$h_{\text{deflection}} = 18.7 \text{ mm}$$

$$50 \text{ mm} \times 18.7 \text{ mm}$$

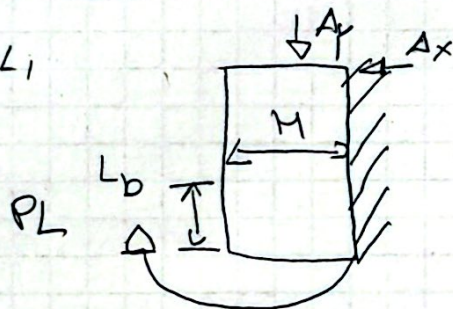
$$b = 50 \quad h = 18.7$$

$$\sigma = \frac{6 (300) (54.6)}{(50) (18.7)^2} = 5.62$$

$$n_{\text{actual}} = \frac{50}{5.62} = 8.89 > 3$$

Feature 2

$$M_1 = PL_1$$



$$M = 55 \text{ mm} \quad b = 2019 \text{ mm}$$

$$M = 16,380 \text{ N}$$

$$\sigma_{\text{max}} = \frac{6M}{bt^2} \quad \sigma \leq \frac{S_y}{n}$$

$$\frac{6M}{bt^2} \leq \frac{S_y}{n}$$

$$t \geq \sqrt{\frac{6Mn}{bS_y}}$$

$$\delta = \frac{MM^2}{2EI} = \frac{MM^2}{2E(bt^{3/12})} = \frac{6MM^2}{Ebt^3}$$

$$\delta = 0.30 = \frac{6MM^2}{Ebt^3}$$

$$t = \sqrt[3]{\frac{6MM^2}{Eb\delta_{\text{max}}}}$$

$$t \geq \sqrt{\frac{6(16,380)(3)}{(50)(44.8)}} \quad t_{\text{stress}} = 11.1 \text{ mm}$$

$$t = 3 \sqrt{\frac{6(16,380)(50)^2}{(1990)(50)(0.30)}} \quad t_{\text{deflection}} = 20.19 \text{ mm}$$

Feature dimensions

Feature 1 =

$$h_{\text{stress}} = 11.1 \text{ mm}$$

$$h_{\text{deflection}} = 18.7 \text{ mm}$$

$$50 \text{ mm} \times 18.7 \text{ mm}$$

Feature 2 =

$$h_{\text{stress}} = 11.1 \text{ mm}$$

$$h_{\text{deflection}} = 20.19 \text{ mm}$$

$$\checkmark L > 55 \text{ mm} \times 20.19$$

Material - ABS

